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course: CSE428

section: 02

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Practice Problem set: 1

Subject

Lecture: 1

Date

Q:1

Picture 1: Absorption.

Picture 2: Emission.

Picture 3: Reflection.

Q:2

Given,

ratio 5:3

Here,

Let,

height $\times$ width = Σ pixels

width (W) = $5x$ (column)

height (H) = $3x$ (Row)

$$15x^2 = 96,0000$$

$$x = \sqrt{\frac{960000}{15}}$$

$$x = 252.98$$

$$\text{width} = 5 \times 252.98 = 1264.9 \approx 1265 \text{ pixels}$$

$$\text{height} = 3 \times 252.98 = 758.94 \approx 759$$

pixels

Size

$$\text{Column} = (1265 \times 8) = 10120 \text{ bits}$$

$$\text{Row} = (759 \times 8) = 6072 \text{ bits}$$

Siraz
Products

Subject :

Date :

8:2

1D: one variable, time/space, $f(t)$ or $f(x)$
ex: sound wave

2D: two variable, grayscale images, $f(x, y)$

3D: Varies with 3 variables, (x, y, t) $f(x, y, t)$
ex: video.

8:4

$f(1, 3) = 4$ [0-based indexing]

8:5 3x3 & 4 bit so, $2^4 = 16$ intensity level.

Range (0 to 15)

9	15	2
15	0	6
8	9	15

Siraz Products

Subject :

Lecture: 2

Date :

Q:1

Given, $f(x, y) = 400$

$$i(x, y) = 700$$

$$r(x, y) = ?$$

We know,

$$f(x, y) = i(x, y) \times r(x, y)$$

$$r(x, y) = \frac{400}{700} = 0.7$$

Range

$\therefore$ Reflectance, $0 \leq r(x, y) \leq 1$

$$f(x, y) : 0 \leq f(x, y) < \infty$$

powers of bright discrimination:

Q:2

Our eyes are better at discriminating

between lower intensity than higher

intensity. weber ratio = $\frac{\Delta I}{I}$

ΔI = small change in intensity,

I = background intensity. At low

intensities weber ratio is small

so, we can detect small change in ~~the~~ brightness.

Subject :

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But X-rays captured dense structures as bright regions & less dense structures as dark region so, which is easier for human eye. That's how it works.

Q:3 Pipeline of Digital Image Acquisition:

↳ captured intensity $0 \leq f(x,y) < \infty$

↳ $f(x,y) = i(x,y) \cdot r(x,y)$

↳ Illumination, $0 \leq i(x,y) < \infty$

↳ Reflectance, $0 \leq r(x,y) \leq 1$

↳ $L_{min} \leq f(x,y) \leq L_{max}$

Dynamic range = $\frac{L_{max}}{L_{min}}$

↳ $[L_{min}, L_{max}]$ is called gray scale

indoor values, $L_{min} \approx 10$, $L_{max} \approx 1000$

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Q:4

$$\text{Dynamic range} = \frac{I_{\max}}{I_{\min}}$$

Img (b) has higher dynamic range.

Because, High dynamic Range allows for better representation of scenes with both very dark & light regions.

It preserves details both shadows & highlights, leading to more realistic that's why HDR is usually desirable for digital img.

Q:5

The camera's sensor consists of pixels. When light falls on a pixel, it generates an electrical charge which is proportional to intensity of the light.

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It convert into a voltage signal.

then ADC digitized it.

Q:6

Two steps: Image Digitization

① Sampling.

② Quantization.

this is important because:

① it convert analog to digital which is helpful for computer.

② It allows storage, manipulation.

Q: 7 Image (a) $\rightarrow$ noise

soln: improve lighting, apply noise reduction.

Image (b) $\rightarrow$ Blurring

soln: use tripod, enable Image Stabilization.

Q: 8 $\&$ use nearest neighbour interpolation.

Down sample:

$$\frac{6+2+2+0}{4} = 2.5 \approx 3$$

$$\frac{1+9+2+4}{4} = 4$$

$$\frac{1+9+2+0}{4} = 3$$

2x2

1	2
2	2

3	4
3	4

$$\frac{2+5+1+2}{4} = 2.5$$

Subject :		Date :	
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up sample: 8×8

6	6	2	2	1	1	9	9
6	6	2	2	1	1	9	9
2	2	0	0	2	2	4	4
2	2	0	0	2	2	4	4
1	1	9	9	9	9	5	5
1	1	9	9	9	9	5	5
2	2	0	0	1	1	2	2
2	2	0	0	1	1	2	2

Q: 9

50/n:

Sensors $\rightarrow$ Retina.

Lens $\rightarrow$ optical . lens

Shutter $\rightarrow$ Iris & pupil

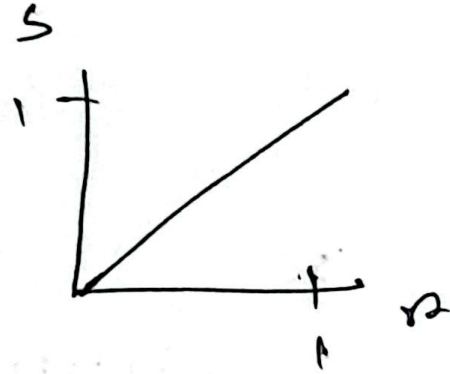
Focusing Region $\rightarrow$ Fovea

Autofocus system / focus Ring $\rightarrow$ Ciliary Body.

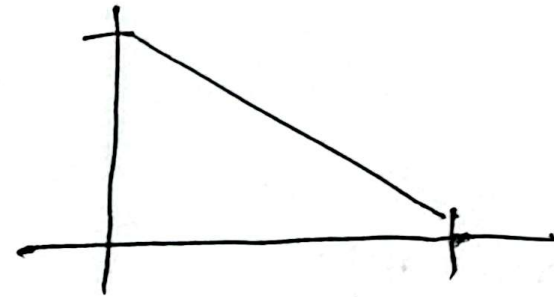
Lecture: 3

Q:1

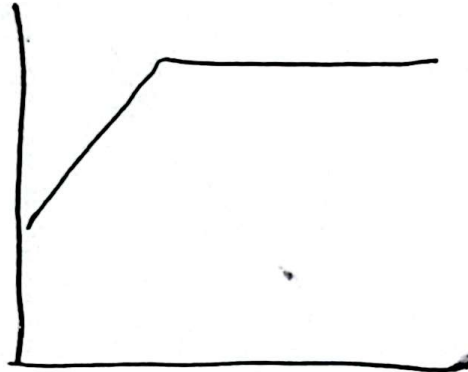
Pic 1:



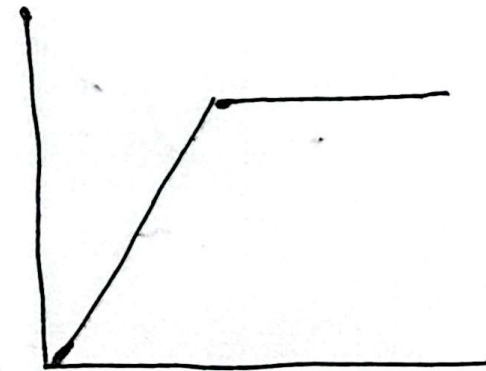
pic 2



Pic:3



pic 4



Q:2 $s = (2 \times r + b) / 255$

i Brightness: Here, $s = 2 \times r + b$, where b is adding, it might saturate high intensity levels. The intensity values increases & the image will be brighter.

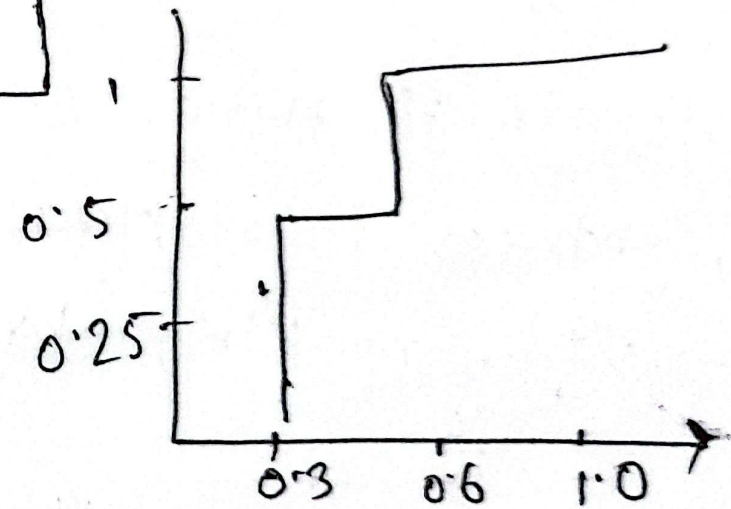
ii Contrast: Here, $s = 2 \times r$, which will increase brightness & contrast. It actually stretches the intensity making brighter pixel brighter.

iii whether there will be any data loss: no value will be lost because it's already clipped in a

Q: 3

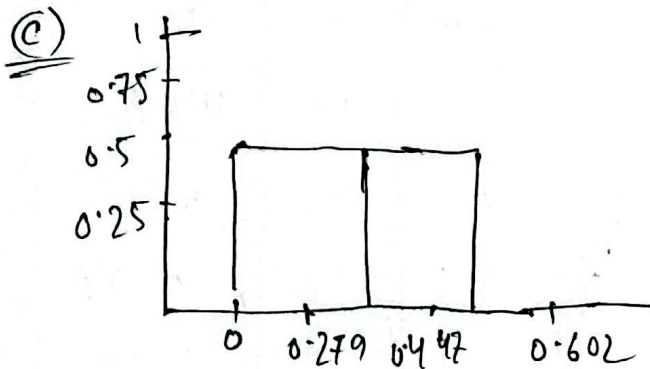
(a)

r_k	$F(r_k)$	$f(r_k)$
0	0	0
0.3	0.5	0.5
0.6	0.5	1
1.0	0	1



3b) $\frac{\max - \min}{\max - \min} = \frac{I - \min}{\max - \min} = 1(I - 0)$

$c_1 = 1, c_2 = 0$



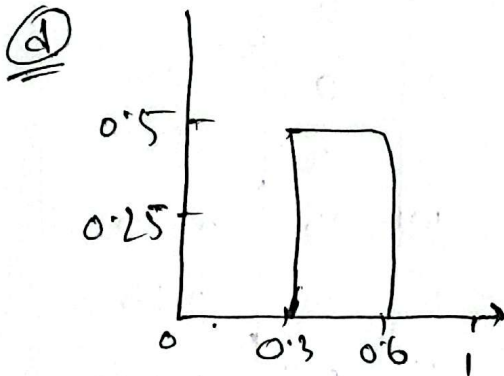
Let,

$I = 0 \Rightarrow \log(1) = 0$

$I = 0 \Rightarrow \log(1 + 3 \times 0.3) = 0.28$

$I = 1 \Rightarrow \log(1 + 3) = 0.60$

$I = 0.6 \Rightarrow \log(1 + 3 \times 0.6) = 0.45$



Q:4

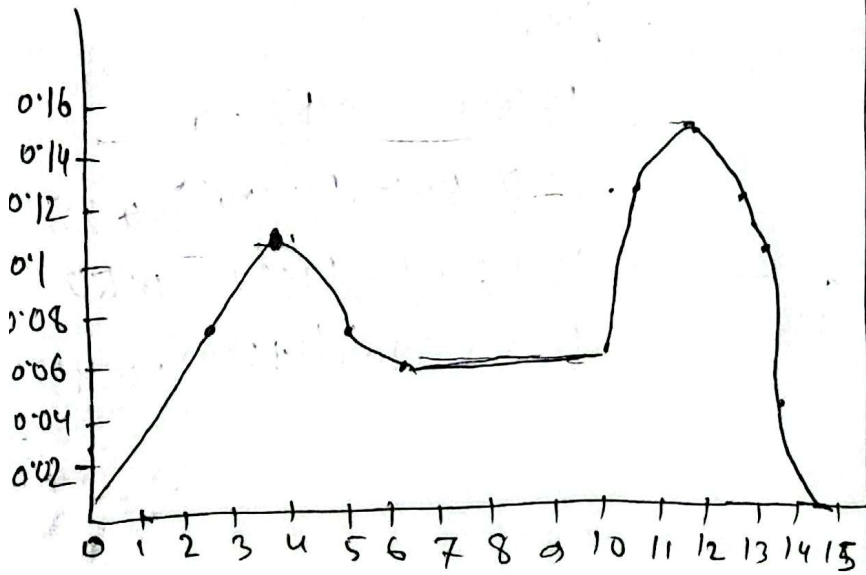
a) $\rightarrow$ e) $\rightarrow$ h) $\rightarrow$ using transformation for slight contrast enhancement. It has balanced histogram.

b) $\rightarrow$ f) $\rightarrow$ i) $\rightarrow$ Using logarithm transformation to compress higher intensities.

c) $\rightarrow$ d) $\rightarrow$ g) $\rightarrow$ Using exponential to stretch lower intensities.

⑤

②

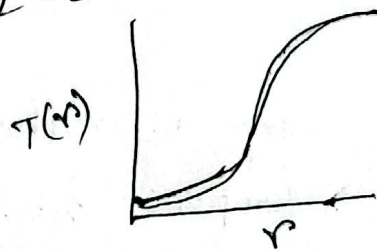


region of interest (3-14)

r_i	n_i	$f(r_i)$
0	0	0
1	0	0
2	0	0
3	13	0.07
4	20	0.11
5	13	0.07
6	11	0.06
7	0	0
8	0	0
9	0	0
10	10	0.06
11	26	0.15
12	28	0.16
13	25	0.14
14	21	0.12
15	9	0.05

⑤ $T(r) = \frac{I-3}{14-3} = 11(I-3)$

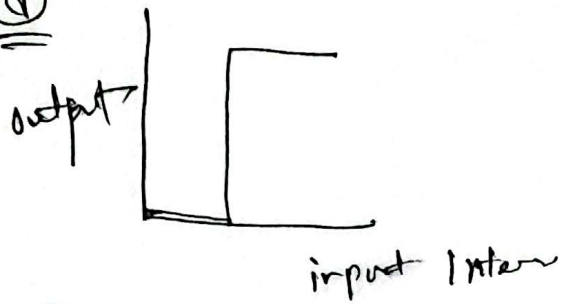
$c_1 = 11, c_2 = 3$



⑥ The gamma value yields the given range $0 < \gamma < 1$

This is because the transformation expands the lower intensity values, resulting in a brighter image with enhanced contrast in the dark area.

①



② a $(x_1, y_1) \rightarrow (9, 2)$
 $(x_2, y_2) \rightarrow (13, 12)$

region (0 → 9)
 $m = \frac{2-0}{9-0} = 2/9$

region (9 → 13)
 $m = \frac{2-0}{9-0} = 2/9$

region (13 → 15)
 $m = \frac{15-14}{15-13} = 0.5$

b region (0 → 9)
 $S = \frac{2r}{9}$

region (9 → 13)
 point: (9, 2)
 $2 = 3 \times 9 + c$
 $\therefore c = -25$

$S = 3r - 25$

region (13 → 15) point (13, 14) a $14 = 13 \times 0.5 + c$
 $c = 7.5$

$S = 0.5r + 7.5$

③ outputs

$0 \rightarrow 0$

$1 \rightarrow \frac{2}{9} \times 1 = 0.22$

$3 \rightarrow \frac{2}{9} \times 3 = 0.67$

$2 \rightarrow \frac{2}{9} \times 2 = 0.44$

$4 \rightarrow \frac{2}{9} \times 4 = 0.89$

$5 \rightarrow \frac{2}{9} \times 5 = 1.11$

$6 \rightarrow \frac{2}{9} \times 6 = 1.33$

$7 \rightarrow \frac{2}{9} \times 7 = 1.56$

$8 \rightarrow \frac{2}{9} \times 8 = 1.78$

$9 \rightarrow \frac{2}{9} \times 9 = 2$

$10 \rightarrow 3 \times 10 - 25 = 5$

$11 \rightarrow 3 \times 11 - 25 = 8$

$12 \rightarrow 3 \times 12 - 25 = 11$

$13 \rightarrow 3 \times 13 - 25 = 14$

$14 \rightarrow 0.5 \times 14 + 7.5 = 14.5$

$15 \rightarrow 0.5 \times 14 + 7.5 = 15$

Lecture: 4

8:1

(a) + (b)

r_k	$h(r_k)$	PDF $f_r(r_k)$	$H(r_k)$	$F(r_k)$ CDF
0	6	$6/16 = 0.375$	6	$6/16 = 0.375$
1	7	$7/16 = 0.438$	$7+6=13$	$13/16 = 0.813$
2	2	$2/16 = 0.125$	$2+13=15$	$15/16 = 0.938$
3	1	$1/16 = 0.063$	$1+15=16$	$16/16 = 1.000$

(b)

(c) $s = T(r_k) = \text{round} \left\{ (L-1) \times f_r(r_k) \right\}$

4 level = 0, 1, 2, 3

$L = 4$

r_k	$F(r_k)$	$s_k = \text{round} \left((4-1) \times F(r_k) \right)$
0	0.375	$\text{round}(1.125) = 1$
1	0.813	$\text{round}(2.439) = 2$
2	0.938	$\text{round}(2.814) = 3$
3	1.000	$\text{round}(3) = 3$

Histogram Equalized.

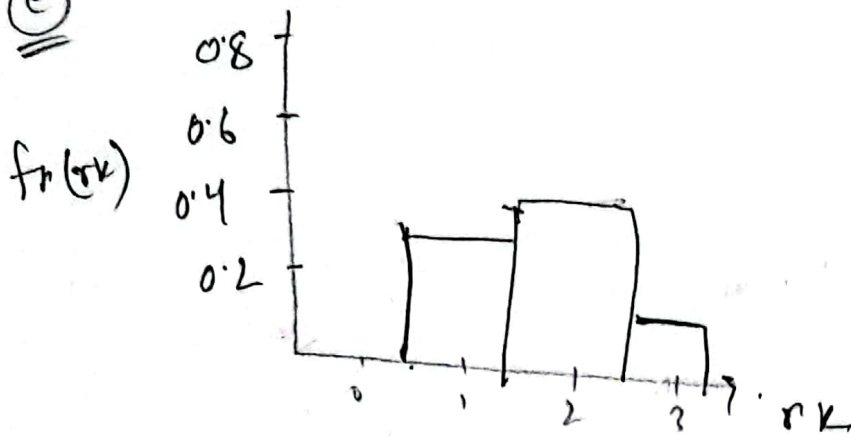
(d) from c we can get equalized, And now replaceable values are

$0 \rightarrow 1$
 $1 \rightarrow 2$
 $2 \rightarrow 3$
 $3 \rightarrow 3$

So, New 4x4 image is

2	3	1	1
2	1	2	3
1	2	1	2
3	1	2	2

②



8.2

②

$$6 + 0 + 41 + 66 + 42 + 23 + x + 7 = 196$$

[14x14]

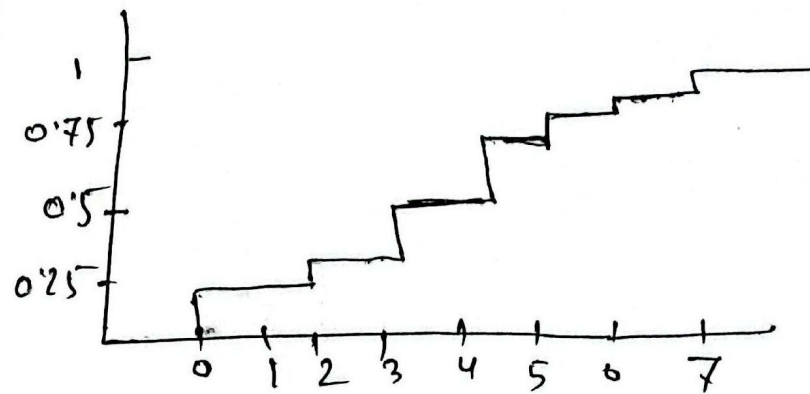
$$\Rightarrow x = 11$$

②

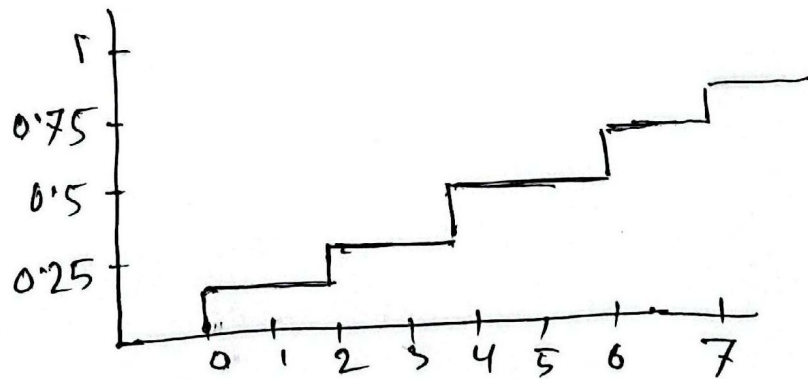
HE Pixels

r	$h(r_k)$	$f(r_k)$ PDF	$H(r_k)$	$F(r_k)$ CDF	$s_r = \text{round}((8-1) \times F(r_k))$
0	6	0.0306	6	0.0306	0
1	0	0.0000	6	0.0306	0
2	41	0.2092	47	0.2398	2
3	66	0.3367	113	0.5765	4
4	42	0.2143	155	0.7908	6
5	23	0.1173	178	0.9081	6
6	11	0.0561	189	0.9642	7
7	7	0.0357	196	1.000	7

2(c)



CDF



Histogram equalized Image [CDF]

After this we see intensity values are better distributed that indicates increament of contrast & thus entropy is increased.

Q:3 a AHE is used. Adaptive Histogram Equalization is used in the picture because it follows the AHE rules like it brighten the object so high like dog's nose portion is visible to high contrast.

b If other techniques like Global Histogram equalization is used it processed uniformly & histogram is equalized globally. And CLAHE is would limit contrast amplification in near-constant, neighbour region. Also it has less noise.

c CLAHE improve over AHE. Because in AHE picture might have higher contrast which cause data or image loss so, for not lossing data or img CLAHE is used. In CLAHE contrast can be controlled so it ensure better picture quality than AHE.